

Classical Machine Learning for Automated Sickle Cell Detection from Peripheral Blood Smear Images: A Ugandan Clinical Dataset Study

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Abstract—Sickle cell disease (SCD) affects approximately 13.3% of the Ugandan population, yet diagnosis remains inaccessible in rural settings due to the requirement for trained laboratory personnel and specialist equipment. This study applies classical machine learning to automate sickle cell detection from peripheral blood smear images using the Tushabe et al. (2024) Ugandan clinical mobile-phone microscopy dataset. Histogram of Oriented Gradients (HOG) feature extraction converts each image into an 8100-dimensional numerical feature vector. Two classifiers are trained and compared: Support Vector Machine (SVM) and Random Forest. Both models are optimised using GridSearchCV with 5-fold cross-validation. SVM achieves the best classical ML performance with 91.98% accuracy and AUC-ROC of 0.9651, while Random Forest achieves higher sensitivity of 96.47%. Results are further compared against deep learning models from a prior study on the same dataset [2], where ResNet-50 achieved 98.57% accuracy. The comparison shows that deep learning significantly outperforms classical ML, with ResNet-50 achieving 98.57% to SVM at 91.98% a viable alternative in low-resource settings where GPU infrastructure for deep learning is unavailable.

Index Terms—Sickle Cell Disease, Machine Learning, HOG Features, SVM, Random Forest, Blood Smear Classification, Uganda, Low-Resource Healthcare

I. INTRODUCTION

Sickle cell disease is a genetic blood disorder caused by a mutation in the HBB gene that produces haemoglobin S, causing red blood cells to deform into a crescent shape under deoxygenated conditions. Uganda carries a high disease burden, with the sickle cell trait estimated at 13.3% nationally and reaching 19.8% in Kampala [1]. An estimated 15,000 to 20,000 children are born with SCD in Uganda each year, many of whom go undiagnosed due to limited access to specialist diagnostic facilities in rural areas.

Current diagnosis depends on laboratory-based peripheral blood smear microscopy, haemoglobin electrophoresis, and sickling tests, all of which require trained personnel and equipment unavailable in most Ugandan district hospitals. Machine learning offers the possibility of automating diagnosis from blood smear images captured using mobile phone cameras on basic microscopes, making screening accessible to non-specialist health workers in rural settings.

This study applies classical machine learning with HOG feature extraction to the Tushabe et al. (2024) Ugandan clinical

microscopy dataset and compares results against deep learning models applied to the same dataset in a prior study [2]. The comparison addresses the question of whether classical ML can produce clinically useful results without the computational cost of deep learning.

II. RELATED WORK

Machine learning has been applied to blood cell analysis with strong results. Acevedo et al. [3] published a dataset of 17,092 annotated peripheral blood cell images and demonstrated automated classification across eight cell types. Deep learning approaches have achieved high accuracy in medical image classification including chest X-ray analysis and blood smear interpretation [4].

For sickle cell detection, prior studies have used datasets collected under controlled laboratory conditions in high-income countries. None used mobile-phone microscopy data from clinical settings in sub-Saharan Africa until the prior study by the same author [2], which applied deep learning to the Tushabe (2024) dataset and achieved 98.57% accuracy with ResNet-50.

HOG features have been widely used for shape recognition since Dalal and Triggs [6] demonstrated their effectiveness. Their ability to capture gradient orientations makes them suited for distinguishing between the morphological characteristics of sickle and normal red blood cells.

III. DATASET

A. Data Sources

Two datasets were combined. The primary dataset is the Tushabe et al. (2024) Ugandan Sickle Cell Microscopy Dataset [1], containing blood smear images collected from patients in Soroti and Kumi districts, Uganda, using mobile phone cameras placed on basic optical microscopes. The dataset includes 422 unlabelled sickle cell positive images and 147 normal images. An additional 422 labelled images were excluded because bounding box annotations were found burned directly into the image pixels, which would cause a classifier to learn annotation artefacts rather than cell morphology.

The supplementary dataset is the BCCD Dataset [7], providing 364 normal peripheral blood smear images captured under

controlled laboratory conditions to supplement the negative class.

B. Dataset Composition

The final combined dataset contains 933 images: 422 positive (45.2%) and 511 negative (54.8%), giving a mild imbalance ratio of 1.21:1. A stratified 80/20 split produced 746 training samples and 187 test samples with class proportions preserved in both sets.

C. Data Challenges

Two data challenges were encountered and resolved. The Mendeley Peripheral Blood Cell dataset [3], originally planned as the normal class source, was found to contain segmented white blood cells rather than red blood cell smear images and was replaced with the BCCD dataset. Additionally, a domain gap exists within the negative class: Tushabe images carry a circular microscope border while BCCD images are rectangular, representing a known limitation.

IV. METHODOLOGY

A. HOG Feature Extraction

HOG feature extraction converts each image into a fixed-length numerical vector by computing gradient orientation histograms in localised image regions. All images were re-sized to 128×128 pixels before extraction to eliminate image dimension as a potential spurious feature. HOG parameters used were 9 orientations, 8×8 pixels per cell, 2×2 cells per block, and L2-Hys block normalisation. Each image produced a feature vector of 8100 values, giving a features matrix of shape 933×8100.

B. Preprocessing Pipeline

The train-test split was performed before any preprocessing to prevent data leakage [8]. A numeric preprocessing pipeline applied StandardScaler to standardise all features to zero mean and unit variance. The scaler was fitted on training data only and used to transform both training and test sets.

C. Classifiers

Support Vector Machine (SVM): SVM finds the optimal hyperplane separating the two classes with maximum margin. The RBF kernel was used with class_weight=balanced to address class imbalance. GridSearchCV optimised C (0.1, 1, 10, 100), kernel (rbf, linear), and gamma (scale, auto) using 5-fold cross-validation with F1 scoring.

Random Forest (RF): Random Forest builds multiple decision trees on random feature subsets and combines predictions by majority vote. GridSearchCV optimised n_estimators (100, 200), max_depth (10, 20, None), min_samples_split (2, 5, 10), and min_samples_leaf (1, 2).

D. Evaluation Metrics

Models were evaluated using accuracy, AUC-ROC, F1 score, sensitivity, specificity, and precision. Overfitting was assessed by comparing training and test accuracy and using 5-fold cross-validation.

V. RESULTS

A. Classical ML Results

The optimised SVM achieved 91.98% test accuracy, AUC-ROC of 0.9651, sensitivity of 92.94%, and specificity of 91.18%, with 15 errors on 187 test images. The optimised Random Forest achieved 87.17% test accuracy, AUC-ROC of 0.9396, and sensitivity of 96.47%, with 24 errors on 187 test images.

Both models showed overfitting. SVM had a training-test accuracy gap of 8.02% and cross-validation mean of 85.12%. Random Forest had a gap of 12.83% and the same cross-validation mean of 85.12%. GridSearchCV partially reduced the overfitting but could not fully resolve it on this small dataset.

B. Comparison with Deep Learning

Table I summarises all five models across both studies. Deep learning results are from [2].

TABLE I
COMPLETE MODEL COMPARISON — CLASSICAL ML VS DEEP LEARNING

Model	Accuracy	AUC-ROC	Sensitivity
SVM (Classical ML)	91.98%	0.9651	92.94%
Random Forest (Classical ML)	87.17%	0.9396	96.47%
Baseline CNN (DL)*	87.86%	0.9588	95.24%
EfficientNet-B0 (DL)*	96.43%	0.9951	95.24%
ResNet-50 (DL)*	98.57%	0.9973	98.41%

*Results from [2]

ResNet-50 achieves the best overall performance. Among classical ML models, SVM performs best. Notably, Random Forest achieves higher sensitivity than both the baseline CNN and EfficientNet-B0, demonstrating that classical ML is not uniformly inferior to deep learning across all clinically relevant metrics.

VI. DISCUSSION

Deep learning outperforms classical ML primarily because it learns features directly from raw images through multiple convolutional layers, automatically discovering discriminative patterns without human design. Transfer learning from ImageNet gives EfficientNet-B0 and ResNet-50 rich visual representations that classical ML cannot replicate with HOG features alone.

However the performance gap between SVM (91.98%) and EfficientNet-B0 (96.43%) is smaller than might be expected, suggesting that HOG features capture meaningful morphological information about cell shape. Random Forest's higher sensitivity of 96.47% compared to EfficientNet-B0's 95.24% is a practically significant finding for clinical screening where minimising false negatives is the priority.

Classical ML has meaningful practical advantages. It requires no GPU infrastructure, trains in minutes rather than hours, and produces results that are more interpretable through feature importance analysis. In low-resource Ugandan health

facilities where GPU hardware is unavailable, SVM with HOG features represents a computationally accessible alternative that still achieves over 90% accuracy.

VII. LIMITATIONS

The dataset of 933 images is small, limiting model generalisation. Both classical ML models show overfitting that was not fully resolved through hyperparameter tuning. The domain gap between Tushabe mobile-phone images and BCCD laboratory images introduces visual inconsistency within the negative class. Comparison with deep learning results uses a different test set size (187 vs 140 samples), which should be considered when interpreting absolute error counts.

VIII. CONCLUSION

This study applied SVM and Random Forest classifiers with HOG feature extraction to the Tushabe (2024) Ugandan clinical microscopy dataset for automated sickle cell detection. SVM achieved 91.98% accuracy and AUC-ROC of 0.9651. Comparison with deep learning models from a prior study shows that ResNet-50 achieves superior overall performance at 98.57% accuracy, but classical ML remains competitive, particularly in sensitivity. Future work should address overfitting through larger datasets, explore additional feature extraction methods, and conduct clinical validation in Ugandan health facilities.

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